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| United States Patent Application Publication | 20250283360 |
| Kind Code | A1 |
| Publication Date | September 11, 2025 |
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## CABINET APPARATUS


### Abstract

A cabinet apparatus having a housing and a door. The housing defines a storage space having a front access opening. The door includes a first panel structure having a first sidewall panel having a first interior surface that partially defines the storage space and a first exterior surface and a tubular flange extending lengthwise along the first sidewall panel and protruding from the first exterior surface. A first hinge component is located on the tubular flange. The door includes a door body and a second hinge component thereon. The first and second hinge components are operably coupled to one another so that the door is pivotably mounted to the housing to be rotatable about a rotational axis between: (1) a closed position in which the door body encloses the front access opening; and (2) an open position in which the storage space is accessible via the front access opening.



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**Family ID:** **90927333**

**Appl. No.:** **19/216936**

**Filed:** **May 23, 2025**

### Foreign Application Priority Data

| | | |
|---|---|---|
| CN | 202211398806.1 | Nov. 09, 2022 |

### Related U.S. Application Data

parent US continuation 17989628 20221117 parent-grant-document US 12320178 child US 19216936

## Publication Classification

**Int. Cl.:** **E05D7/081** (20060101); **A47B67/02** (20060101); **E05D11/06** (20060101)

**U.S. Cl.:**

CPC **E05D7/081** (20130101); **A47B67/02** (20130101); **E05D11/06** (20130101); E05Y2201/224 (20130101); E05Y2600/40 (20130101); E05Y2600/622 (20130101); E05Y2900/20 (20130101)

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## Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/989,628 (Docket No. 010222-23232A) filed on Nov. 17, 2022, which is hereby incorporated by reference in its entirety, and which claims the benefit of Chinese National patent application No. 202211398806.1, filed on Nov. 9, 2022, the contents thereof hereby incorporated by reference herein in its entirety, for all purposes.

BACKGROUND
[0002] Cabinets are used throughout the home and in certain office environments to store items out of sight but in an easily accessible location. For example, medicine cabinets are typically hung in a bathroom and include a mirrored door. People typically store items related to personal hygiene in medicine cabinets, such as deodorant, toothpaste, toothbrushes, hairbrushes, and medication. Some users may choose to store cosmetics, hair curlers, hair dryers, and hair removal/shaving materials in the cabinet. Users may take advantage of the cabinet's mirrored door to apply cosmetics, apply medication, dry/curl/brush hair, and/or shave/remove hair.
SUMMARY
[0003] The present disclosure may be directed to a cabinet apparatus having a housing and a door. The housing may define a storage space having a front access opening. The door may include a first panel structure having a first sidewall panel having a first interior surface that partially defines the storage space and a first exterior surface. The door may also have a tubular flange extending lengthwise along the first sidewall panel and protruding from the first exterior surface. A first hinge component may be located on the tubular flange. The door may include a door body and a second hinge component on the door body. The first and second hinge components may be operably coupled to one another so that the door is pivotably mounted to the housing to be rotatable about a rotational axis between: (1) a closed position in which the door body encloses the front access opening; and (2) an open position in which the storage space is accessible via the front access opening.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0004] The present invention will become more fully understood from the detailed description and the accompanying drawings, wherein:
[0005] FIG. **1** is a front perspective view of an example cabinet assembly having a housing and a door with the door in a closed position;
[0006] FIG. **2** is a front perspective view of an example cabinet assembly having a housing and a door with the door in a partially opened position;

[0007] FIG. **3** is a partially exploded front perspective view of an example cabinet assembly having a housing and a door shown detached from one another
[0008] FIG. **4**A is a front perspective view of an example housing of a cabinet apparatus;
[0009] FIG. **4**B is an exploded front perspective view of an example housing of a cabinet apparatus;
[0010] FIG. **5**A is a partially exploded rear perspective view of an example cabinet assembly;
[0011] FIG. **5**B is a close-up view of area VB of FIG. **5**A;
[0012] FIG. **6** is a front perspective view of an example door of a cabinet apparatus;
[0013] FIG. **7** is an exemplary close-up view of area VII of FIG. **3**;
[0014] FIG. **8** is a front perspective view of an exemplary cabinet apparatus shown flush mounted to a wall;
[0015] FIG. **9** is a front perspective view of an exemplary cabinet apparatus shown recess mounted to a wall;
[0016] FIG. **10** is a top plan view of an exemplary cabinet apparatus shown recess mounted to a wall with a door of the cabinet apparatus in a closed position;
[0017] FIG. **11** is a close-up view of area XI of FIG. **10**;
[0018] FIG. **12** is a top plan view of an exemplary cabinet apparatus shown recess mounted to a wall with a door of the cabinet apparatus in an open position;
[0019] FIG. **13** is a close-up view of area XIII of FIG. **12**;
[0020] FIG. **14** is a front perspective view of an example cabinet assembly having a housing and a door with the door in a closed position; and
[0021] FIG. **15** is a front perspective view of an example cabinet assembly having a housing and a door with the door in a partially opened position.

DETAILED DESCRIPTION

[0022] The following description of the is merely exemplary in nature and is in no way intended to limit the invention, its application, or uses.

[0023] The description of illustrative embodiments according to principles of the present invention is intended to be read in connection with the accompanying drawings, which are to be considered part of the entire written description. In the description of embodiments of the invention disclosed herein, any reference to direction or orientation is merely intended for convenience of description and is not intended in any way to limit the scope of the present invention. Relative terms such as “lower,” “upper,” “horizontal,” “vertical,” “above,” “below,” “up,” “down,” “top,” and “bottom” as well as derivatives thereof (e.g., “horizontally,” “downwardly,” “upwardly,” etc.) should be construed to refer to the orientation as then described or as shown in the drawing under discussion. These relative terms are for convenience of description only and do not require that the apparatus be constructed or operated in a particular orientation unless explicitly indicated as such. Terms such as “attached,” “affixed,” “connected,” “coupled,” “interconnected,” and similar refer to a relationship wherein structures are secured or attached to one another either directly or indirectly through intervening structures, as well as both movable or rigid attachments or relationships, unless expressly described otherwise. Moreover, the features and benefits of the invention are illustrated by reference to the exemplified embodiments. Accordingly, the invention expressly should not be limited to such exemplary embodiments illustrating some possible non-limiting combination of features that may exist alone or in other combinations of features; the scope of the invention being defined by the claims appended hereto.

[0024] Referring to FIGS. **1** and **2**, a cabinet apparatus **100** will be described. The cabinet apparatus **100** generally includes a housing **200** and a door **300**. The door **300** may be pivotably coupled to the housing **200** between a closed position as shown in FIG. **1** and an open position as shown in FIG. **2**. The door **300** may be pivotably mounted to the housing **200** to be rotatable about a rotational axis A-A between the open and closed positions. The housing **200** may have a compartment body **201** that defines a storage space **202** and a flange **260** protruding outward from the compartment body **201**. The storage space **202** may have a front access opening **299** which is an opening through which users may insert and remove items from the storage space **202**. The flange **260** may circumscribe the front access opening **299**.

[0025] The storage space **202** may be an interior cavity of the housing **200** within which a user may store various objects, such as for example without limitation medication, toothbrush, deodorant, hair care supplies, or really any other desired item. The cabinet apparatus **100** may include one or more shelves **203** located within the storage space **202** upon which a user may place any of the items that are stored within the storage space **202**. The shelves **203** may be spaced apart from one another by varying distances which may be predetermined by a manufacturer or selectable by a consumer or end user. While there are three or four shelves **203** depicted in the drawings, there may be more or less than three or four shelves **203**. The shelves **203** may be supported within the storage space **202** by clips **298** attached to interior surfaces of the compartment body **201**.

[0026] The door **300** may have an exterior surface **301** and an interior surface **302**. When the door **300** is in the closed position, the interior surface **302** may face the storage space **202** and the exterior surface **301** may face away from the storage space **202**. The door **300** may close the front access opening **299** when in the closed state. The exterior surface **301** of the door **300** may be formed by or include a mirror **303** to allow a user to view themselves as is conventional for medicine style cabinets. Alternatively, the exterior surface **301** of the door **300** may not be mirrored, but may instead be a desirable color, texture, pattern, or the like to achieve a desired aesthetic for a particular space within which the cabinet apparatus **100** is to be hung. The interior surface **302** of the door **300** may also be formed by or include a mirror **309** to allow a user to view themselves when the door **300** is in the open position. Alternatively, the interior surface **302** may not be mirrored. The door **300** may be rectangular with rounded corners as shown, or may take on other shapes and sizes, including being rectangular or square with sharp or rounded corners, triangular, oval, circular, or other polygonal shapes with either sharp or rounded corners. The housing **200** may be rectangular regardless of the shape of the door **300**. Alternatively, the housing **200** may also take on other shapes including circular, oval, square, or other polygonal shapes.

[0027] The housing **200** may have a front face **211** which surrounds the front access opening **299** and which is exposed when the door **300** is in the open position. The front face **211** of the housing **200** may be formed by a front surface of the flange **260**. The housing **200** may have one or more magnetic elements **212** located on the front face **211**. The one or more magnetic elements **212** may protrude from the front face **211**. Alternatively, the one or more magnetic elements **212** may nest within a recess in the front face **211**. The one or more magnetic elements **212** may include a magnetic member that is located within a silicone, rubber, foam, or other material which forms a housing that contains the one or more magnetic elements **212**. Thus, the one or more magnetic elements **212** when housed within a silicone, rubber, foam, or other material may form a bumper element to prevent the door **300** from slamming against the housing **200** when being altered from the open position to the closed position. Alternatively, the magnetic elements **212** may be omitted.

[0028] The interior surface **302** of the door **300** may have one or more pockets **304**, although such pockets **304** could be omitted. The door **300** may include magnet elements (not shown) located within the one or more pockets **304**. The one or more pockets **304** may be aligned with the one or more magnetic elements **212** when the door **300** is in the closed state. As such, the one or more magnetic elements **212** of the housing **200** may nest within the one or more pockets **304** of the door **300** when the door **300** is in the closed position. The magnetic elements **212** of the housing **200** may magnetically attract to magnetic elements in the pockets **304** of the door **300** to secure the door **300** in the closed position.

[0029] Referring to FIG. **3**, the cabinet apparatus **100** is illustrated with the door **300** detached from the housing **200**. The housing **200** may include a first hinge component **280**. The door **300** may include a second hinge component **330**. The first and second hinge components **280**, **330** may be operably coupled together to enable the pivoting movement of the door **300** between the open and closed positions. The cabinet apparatus **100** may further include hardware components for facilitating the attachment between the first and second hinge components **280**, **330**. The hardware components may include one or more hinge pins **400**, one or more washers **401**, and one or more connector members **402** that facilitate the attachment between the first and second hinge components **280**, **330** while permitting the door **300** to rotate/pivot relative to the housing **200**. The first and second hinge components **280**, **330** may be operably coupled to one another so that the door **300** is pivotably mounted to the housing **200** to be rotatable about the rotational axis A-A between: (1) the closed position in which the door **300** closes the front access opening **299** of the storage space **202**, as shown in FIG. **1**; and (2) the open position in which the storage space is accessible via the front access opening **299**, as shown in FIG. **2**.

[0030] Referring to FIGS. **4**A and **4**B, the housing **200** will be described in greater detail. The housing **200** may have a rear panel **204** and a plurality of panel structures **205***a-d* that may be connected to and extend from the rear panel **204** to define the storage space **202**. The rear panel **204** may be a flat panel structure having a flat interior surface **213** and a flat exterior surface **214**. Alternatively, the interior and/or exterior surfaces **213**, **214** of the rear panel **204** may have some contour rather than being entirely flat and planar. The interior surface **213** of the rear panel **204** may face and define a rearmost boundary of the storage space **202**.

[0031] The panel structures **205***a-d* may include a top panel structure **205***a*, a bottom panel structure **205***b*, a left panel structure **205***c*, and a right panel structure **205***d*. Each of the panel structures **205***a-d* may be coupled to the rear panel **204** to form the housing **200** and to define the storage space **202**. Various clips, fasteners, screws, brackets, and the like may be used to facilitate the attachment of the panel structures **205***a-d* to the rear panel **204**. Each of the panel structures **205***a-d* may extend from the interior surface **213** of the rear panel **204** to collectively define the storage space **202**. As discussed below, the compartment body **201** may include the rear panel **204** and parts of each of the panel structures **205***a-d*, while other parts of the panel structures **205***a-d* may form the flange **260**.

[0032] The top panel structure **205***a* may include a top panel **220** that forms an upper boundary of the storage space **202**, a top flange portion **221**, and a hanging bracket **222**. The top panel **220** may have a front edge **223**, a rear edge **224**, an interior surface **225**, and an exterior surface **226**. The top panel **220** may be a flat panel structure such that the interior and exterior surfaces **225**, **226** are planar. The top flange portion **221** may extend upwardly from the exterior surface **226** of the top panel **220** along the front edge **223** of the top panel **220**. The top flange portion **221** may be L-shaped, such that it may include a vertical wall portion **227** that extends upwardly from the exterior surface **226** of the top panel **220** and a horizontal wall portion **228** that extends horizontally and rearwardly from a distal end of the vertical wall portion **227** in a direction towards the rear edge **224** and towards the rear panel **204**. Thus, the top flange portion **221** may form an open channel. In an alternative embodiment, the top flange portion **221** may include another vertical wall extending from the distal end of the horizontal wall portion **228** to the exterior surface **231** of the top panel **220**. The top panel structure **205***a* may be formed as a single monolithic component via an extrusion process.

[0033] The hanging bracket **222** may extend upwardly from the exterior surface **224** of the top panel **220** along the rear edge **224** of the top panel **222**. The hanging bracket **222** may be used when the cabinet apparatus **100** is flush-mounted to a wall, whereas the hanging bracket **222** may not be used when the cabinet apparatus **100** is recess mounted to a wall. When the cabinet apparatus **100** is flush-mounted to a wall, cleats or the like may be mounted to the wall and then the hanging bracket **222** may engage the cleats to hang or mount the cabinet apparatus **100** to the wall. The cabinet apparatus **100** may also include separate brackets **215** with apertures for further securing the cabinet apparatus **100** to the wall when flush mounted as described herein. The hanging bracket **213** may extend upwardly from the top boundary wall **220** and may be located adjacent to a rear edge of the top panel structure **205***a* which is adjacent to the rear panel **204**. The hanging bracket

**213** may extend along the full length of the top panel structure **205***a*, although alternatively it may extend along a portion of the length of the top panel structure **205***a*.

[0034] The bottom panel structure **205***b* may include a bottom panel **229** having an interior surface **230**, an exterior surface **231**, a front edge **232**, and a rear edge **233**. The bottom panel **229** may be a flat panel structure such that the interior and exterior surfaces **230**, **231** are planar. The bottom panel **229** may form a lower boundary of the storage space **202**. In particular, the interior surface **230** of the bottom panel **229** may form the lower boundary of the storage space **202**. The bottom panel structure **205***b* may further comprise a bottom flange portion **209**. The bottom flange portion **209** may be located along the front edge **232** of the bottom panel **229**. In particular, the bottom flange portion **209** may be L-shaped such that it may include a vertical wall portion **234** that extends downwardly from the exterior surface **230** of the bottom panel **229** along the front edge **232** and a horizontal wall portion **235** that extends rearwardly from the distal end of the vertical wall portion **234** in a direction towards the rear edge **233** of the bottom panel **229** and towards the rear panel **204**. Thus, the bottom flange portion **209** may form an open channel. In an alternative embodiment, the bottom flange portion **209** may include another vertical wall extending from the distal end of the horizontal wall portion **235** to the exterior surface **231** of the bottom panel **229**. The bottom panel structure **205***b* may be formed as an integral monolithic component via an extrusion process.

[0035] The left panel structure **205***c* (which may be referred to in the specification and/or claims as a first panel structure) includes a first sidewall panel **236** having an interior surface **237** and an exterior surface **238**. The first sidewall panel **236** may extend lengthwise and be elongated between a top edge **239** and a bottom edge **240**. The first sidewall panel **236** may have a front edge **241** and a rear edge **242**. The interior and exterior surfaces **237**, **238** of the first sidewall panel **236** may be generally planar. The interior surface **237** of the left sidewall panel **236** may define or bound a left side of the storage space **202**. The left panel structure **205***c* may further include a left flange portion **243**. The left flange portion **243** may be a tubular flange extending lengthwise along the first sidewall panel **236** from the top edge **239** to the bottom edge **240**. The left flange portion **243** may be located adjacent to the front edge **241** of the first sidewall panel **236**. The left flange portion **243** may protrude from the exterior surface **238** of the first sidewall panel **236** adjacent to the front edge **241** of the first sidewall panel **236**.

[0036] The left flange portion **243** is described herein as being tubular, but this does not limit the left flange portion **243** to being any particular shape. In the example shown, the left flange portion **243** is square or rectangular and tubular, meaning it defines a hollow interior. However, alternatively the left flange portion **243** may take on other shapes, such as being circular, triangular, or other polygonal shapes. The structural details of the left flange portion **243** will be described in greater detail below with reference to FIGS. **5**A and **5**B.

[0037] The left panel structure **205***c* may further include the first hinge component **280**. The first hinge component **280** may be located on the left flange portion (or tubular flange) **243** of the left panel structure **205***c*. The first hinge component **280** may include an upper first hinge knuckle **281** protruding from a front-outside corner of the left flange portion **243** adjacent to the top edge **239** of the first sidewall panel **236** and a lower first hinge knuckle **282** protruding from the front-outside corner of the left flange portion **243** adjacent to the bottom edge **240** of the first sidewall panel **236**. Thus, the first hinge component **280** may include a pair of the first hinge knuckles **281**, **282** (including the upper and lower first hinge knuckles) that are spaced apart along the length of the left flange portion **243** of the left panel structure **205***c*. Alternatively, a single first hinge knuckle or more than two first hinge knuckles could be used. Additional details about the first hinge knuckles **281**, **282** may be provided below with reference to FIGS. **5**A and **5**B.

[0038] The left panel structure **205***c* may be a singular monolithic component which includes the first sidewall panel **236**, the left flange portion **243**, and the first hinge knuckles **231**, **231** as integral parts of a singular component. The left panel structure **205***c* may be formed via an

extrusion process.
[0039] As noted above, the left flange portion **243** is tubular, meaning it is not just an L-shaped flange, but rather is a fully enclosed flange with a square or rectangular transverse cross-sectional area. Thus, the left flange portion **243** is more robust and rigid than if it were just an L-shaped flange. This added robustness by forming the left flange portion **243** with a tubular shape enables the first hinge component **280** to be located on the left flange portion **243** and to be formed integrally with the left flange portion **243**. Thus, rather than having to affix a separate metal or plastic hinge to the housing **200**, the hinge (or the first hinge component **280** there) is able to be formed integrally with the left panel structure **205***c*.
[0040] The housing **200** may also include a first cover member **244** that is configured to be detachably coupled to the left panel structure **205***c* to at least partially cover the exterior surface **238** of the first sidewall panel **236**. The first cover member **244** may cover portions of the exterior surface **238** of the first sidewall panel **236** that would otherwise be exposed. In particular, the first cover member **244** may cover portions of the exterior surface **238** of the first sidewall panel **236** from which the left flange portion **243** does not protrude. When coupled to the left panel structure **205***c*, the first cover member **244** may sit flush with an exterior surface of the left flange portion **243** to create a seamless exterior appearance. The first cover member **244** may be used when the cabinet apparatus **100** is flush mounted to a wall, but the first cover member **244** may be omitted when the cabinet apparatus **100** is recess mounted to a wall. The housing **200** may include one or more bracket members **245** to facilitate the coupling of the cover member **244** to the left panel structure **205***c*. The one or more bracket members **245** may be coupled to the first sidewall panel **236** with fasteners such as screws, and then coupled to the first cover member **244** via mechanical interaction of the parts (sliding engagement, mating, snap-fit, friction fit, or the like). Alternatively, the one or more bracket members **245** may also be coupled to the first cover member **244** with screws or fasteners of other types.
[0041] The right panel structure **205***d* is essentially identical to the left panel structure **205***c*, except the right panel structure **205***d* may not include the hinge component. Alternatively, the right panel structure **205***d* could include the hinge component **280** while the left panel structure **205***c* does not include the hinge component **280**. This may be dependent on the direction upon which the door **300** is intended to pivot between the open/closed positions.
[0042] The right panel structure **205***d* (which may be referred to in the specification and/or claims as a second panel structure) includes a second sidewall panel **246** having an interior surface **247** and an exterior surface **248**. The second sidewall panel **246** may extend lengthwise and be elongated between a top edge **249** and a bottom edge **250**. The second sidewall panel **246** may have a front edge **251** and a rear edge **252**. The interior and exterior surfaces **247**, **248** of the second sidewall panel **246** may be generally planar. The interior surface **247** of the second sidewall panel **246** may define or bound a right side of the storage space **202**. The second panel structure **205***d* may further include a right flange portion **253**. The right flange portion **253** may be a tubular flange extending lengthwise along the second sidewall panel **246** from the top edge **249** to the bottom edge **250**. The right flange portion **253** may be located adjacent to the front edge **251** of the second sidewall panel **246**. The right flange portion **253** may protrude from the exterior surface **248** of the second sidewall panel **246** adjacent to the front edge **251** of the second sidewall panel **246**.
[0043] The right flange portion **253** is described herein as being tubular, but this does not limit the right flange portion **253** to being any particular shape. In the example shown, the right flange portion **253** is square or rectangular and tubular, meaning it defines a hollow interior. However, alternatively the right flange portion **253** may take on other shapes, such as being circular, triangular, or other polygonal shapes. The structural details of the right flange portion **253** may be identical to the left flange portion **243**, and thus the description of the left flange portion **243** below with reference to FIGS. **5**A and **5**B may be entirely applicable to the right flange portion **253**.
[0044] The right panel structure **205***d* may be a singular monolithic component which includes the

second sidewall panel **246** and the right flange portion **253**. The right panel structure **205***d* may be formed via an extrusion process.

[0045] The housing **200** may also include a second cover member **254** that is configured to be detachably coupled to the right panel structure **205***d* to at least partially cover the exterior surface **248** of the second sidewall panel **246**. The second cover member **254** may cover portions of the exterior surface **248** of the second sidewall panel **246** that would otherwise be exposed. In particular, the second cover member **244** may cover portions of the exterior surface **248** of the second sidewall panel **246** from which the right flange portion **253** does not protrude. When coupled to the right panel structure **205***d*, the second cover member **254** may sit flush with an exterior surface of the right flange portion **253** to create a seamless exterior appearance. The second cover member **254** may be used when the cabinet apparatus **100** is flush mounted to a wall, but the second cover member **254** may be omitted when the cabinet apparatus **100** is recess mounted to a wall. The housing **200** may include one or more bracket members **255** to facilitate the coupling of the second cover member **254** to the right panel structure **205***d*. The one or more bracket members **255** may be coupled to the second sidewall panel **246** with fasteners such as screws, and then coupled to the second cover member **254** via mechanical interaction of the parts (sliding engagement, mating, snap-fit, friction fit, or the like). Alternatively, the one or more bracket members **255** may also be coupled to the second cover member **254** with screws.

[0046] Various additional hardware components such as brackets, fasteners, screws, clips, and the like are shown in FIG. **4**B and may be used to couple the various components described above together and/or to facilitate the mounting of the assembled cabinet apparatus **100** to a wall. These hardware components are generally self-explanatory and a detailed discussion thereof is not provided here.

[0047] In accordance with the disclosure set forth herein, the top panel **220**, the bottom panel **229**, the first sidewall panel **236**, and the second sidewall panel **246** collectively form the compartment body **201** of the housing **200**, which defines the storage space **202**. That is, the interior surfaces **225**, **230**, **237**, **247** of the top, bottom, first sidewall, and second sidewall panels **220**, **229**, **236**, **246** may collectively form the boundary of the storage space **202**. Furthermore, the top flange portion **227**, the bottom flange portion **209**, the left flange portion **243**, and the right flange portion **253** may collectively form the flange **260** of the housing **200**. The left and right flange portions **243**, **254** may be tubular structures and the top and bottom flange portions **227**, **209** may be open channel structures (such as formed by the L-shape as described above). As discussed further below, the flange **260** is configured to abut against a surface of a wall when the cabinet apparatus **100** is recess-mounted into the wall. That is, the flange **260** provides a contact surface which will abut against the wall to ensure that the cabinet apparatus **100** is recessed a proper distance into the wall when recess-mounted. The flange **260** may surround the front access opening **299** of the housing **200**.

[0048] Referring to FIGS. **5**A and **5**B, the left (or first) panel structure **205***c*, and more specifically the left flange portion **243** and first hinge component **280** thereof, will be further described. The left flange portion (or tubular flange) **243** may include a front wall **256**, an outer wall **257**, a rear wall **258**, and an inner wall **259** that collectively define a tubular cavity **261**. The tubular cavity **261** may have a rectangular or square transverse cross-sectional profile. In alternative embodiments, the tubular cavity **261** may have a differently shaped cross-sectional profile, such as being other polygonal shapes or circular. The first sidewall panel **236** may have a front portion which forms the inner wall **259** of the left flange portion **243**. That is, the first sidewall panel **236** may be flush and seamless with the inner wall **259** of the left flange portion **243** because in essence the first sidewall panel **236** forms the inner wall **259**. The front and rear walls **256**, **258** may extend perpendicularly from the inner wall **259** in the same direction, and the outer wall **257** may extend between the opposing ends of the front and rear walls **256**, **258** relative to the inner wall **259** so that the outer wall **257** may be parallel to the inner wall **259**. The front wall **256** may form a portion of the front face **211** of the housing **200** when the housing **200** is assembled. The left flange portion or tubular flange **243** has a front-outside corner **262** at the intersection of the front wall **256** and the outer wall **257**. As used herein, the front-outside corner **262** may include the edge where the front and outer walls **256**, **257** intersect as well as a small portion of the front and outer walls **256**, **257** which is immediately adjacent to the intersection edge. The first hinge component **280** may be located at and protrude from the front-outside corner **262** of the left flange portion **243**. More specifically, the first hinge component **280** may protrude from the outer wall **257** at a position that is immediately adjacent to location at which the front and outer walls **256**, **257** intersect (i.e., the first hinge component **280** may protrude from the outer wall **257** at a position that is adjacent to the front wall **256**).

[0049] The outer wall **257** may extend rearwardly beyond the rear wall **258** such that the portion of the outer wall **257** that extends beyond the rear wall **258** forms an abutment lip **263**. The abutment lip **263** may protrude from an exterior surface of the rear wall **258** which faces away from the tubular cavity **261**. The abutment lip **263** may abut against a wall surface when the cabinet apparatus **100** is recess-mounted into the wall, as described further below. Alternatively, the abutment lip **263** could be omitted and in such alternatives examples the exterior surface of the rear wall **258** may abut against the wall surface when the cabinet apparatus **100** is recess-mounted into the wall.

[0050] The abutment lip **263** terminates in a distal end **266** which is the end that is configured to abut against the wall when the cabinet apparatus **100** is recess-mounted into a wall. The distal end **266** may have an increased thickness relative to a remainder of the abutment lip **263** to enhance the contact surface area between the abutment lip **263** and the wall. Although not shown in these figures, when the cover member **244** is coupled to the left panel structure **205***c*, one of the edges of the cover member **244** abuts directly against the distal end **266** of the abutment lip **263**. Furthermore, the front surface of the cover member **244** is flush with the exterior surface of the outer wall **257**. This is shown, for example, in FIG. **4**A with reference to the second cover member **254** and its interaction with the right flange portion **253** of the second sidewall panel **246**. That is, the flush interaction between the second cover member **254** and the right flange portion **253** is shown in FIG. **4**A, and the same flush interaction may be formed between the first cover member **244** and the left flange portion **243**.

[0051] The first hinge component **280** may include the upper first hinge knuckle **281** and the lower first hinge knuckle **282**. The upper first hinge knuckle **281** will be described here, and it should be understood that the lower first hinge knuckle **282** is identical to the upper first hinge knuckle **282** and thus the description of the upper first hinge knuckle **281** is entirely applicable to the lower first hinge knuckle **282**. Furthermore, in alternative examples there may be just one first hinge knuckle rather than an upper one and a lower one that are spaced apart.=

[0052] The first hinge component **280** may include a first rib **283** protruding from the left flange portion **243**. The first rib **283** may protrude from the left flange portion **243**, and more specifically from the outer wall **257** thereof, at or adjacent to the front-outside corner **262** as described above. The upper first hinge knuckle **281** may be attached to and extend or protrude from a distal end of the first rib **283** which is spaced from the left flange portion **243**. The upper first hinge knuckle **281** may include a cylindrical body portion **284** that defines a hollow interior **285** extending vertically through the cylindrical body portion **284**.

[0053] A stop lip **264** may protrude from an exterior surface of the outer wall **257** of the left flange portion **243**. The stop lip **264** may extend the full length between the upper and lower first hinge knuckles **281**, **282**. Alternatively, the stop lip **264** may extend intermittently or only part of the way between the upper and lower first hinge knuckles **281**, **282**. The stop lip **264** may be formed by at least a portion of the front wall **256** extending beyond the exterior surface of the outer wall **257**. Alternatively, the stop lip **264** may be a separate tab feature that protrudes from the exterior surface of the outer wall **257** at a location adjacent to the front-outside corner **262**. An exterior surface of the stop lip **264** may be flush with an exterior surface of the front wall **256**.

[0054] As previously described, the cabinet apparatus **100** may include one or more hinge pins **400**, one or more washers **401**, and one or more connector members **402** that facilitate the pivoting/rotating attachment between the first and second hinge components **280**, **330**. More specifically, the one or more hinge pins **400** may include a first hinge pin **400***a* that may be used to pivotably couple the upper first hinge knuckle **281** to the second hinge component **330** of the door **300** and a second hinge pin **400***b* that may be used to pivotably couple the lower first hinge knuckle **282** to the second hinge component **330** of the door **300**. There may be a washer **401** and/or a connector member **402** associated with each of the hinge pins **400***a*, **400***b*.

[0055] Referring to FIG. **6**, the door **300** will be described. The door **300** may include a door body **305** having an exterior surface (or front surface) **306** and an interior surface (or rear surface) **307**. The exterior surface **306** may form the exterior surface **301** of the door **300** and the interior surface **307** may form the interior surface **302** of the door **300**. Alternatively, a mirror may be affixed to one or both of the exterior and interior surfaces **306**, **307** of the door body **305** such that the mirror may form the exterior and interior surfaces **301**, **302** of the door **300**. The second hinge component **330** may be located on the door body **301**. The second hinge component **330** may comprise a second hinge knuckle **331** that protrudes from the interior surface **307** of the door body **305** of the door body **305**. The second hinge knuckle **331** may be located adjacent to an edge of the door body **305** which is located adjacent to the housing **200** when the door **300** is coupled to the housing **200** to form the cabinet apparatus **100**.

[0056] The second hinge knuckle **331** may include a cylindrical body **332** that is elongated between a top end **333** and a bottom end **334**. The cylindrical body **332** may define a hollow interior **336**, or may be hollow along top and bottom portions thereof. The second hinge component **330** may also include a hinge stop **335**. The hinge stop **330** may be a tab feature that protrudes from an exterior surface of the cylindrical body **332** of the second hinge knuckle **331**. The hinge stop **335** may extend along the full length of the cylindrical body **332**, or along only a portion of the length of the cylindrical body **332**. The hinge stop **335** may protrude from the cylindrical body **332** in a direction that is generally perpendicular to the interior surface **307** of the door body **305** and at a position along the cylindrical body **332** which is closest to the edge of the door body **305** that the cylindrical body **332** is positioned nearest to. While the hinge stop **335** is shown and described as being formed as part of the second hinge component **330**, in other embodiments the hinge stop **335** maybe formed as part of the first hinge component **280** of the housing **200**.

[0057] Referring to FIG. **7**, the coupling of the first and second hinge components **280**, **330** will be described. The second hinge knuckle **331** of the second hinge component **330** of the door **300** may be positioned between the upper and lower first hinge knuckles **281**, **282** of the first hinge component **280** of the housing **200**. Then, the first hinge pin **400***a* may be inserted through the passageways in the upper hinge knuckle **281** and the second hinge knuckle **331** and the second hinge pin **400***b* may be inserted through the passageways in the lower hinge knuckle **282** and the second hinge knuckle **331**. The washer element **401** and connector members **402** may further assist with the connection of the first and second hinge components **280**, **330** will permitting the pivoting movement as described herein. Once the elements described here are assembled, the door **300** is pivotably coupled to the housing **200** and may be rotated or pivoted between the open and closed positions.

[0058] FIG. **8** illustrates the cabinet apparatus **100** being flush-mounted to a wall **500**. When the cabinet apparatus **100** is flush-mounted to the wall **500**, the first and second cover members **244**, **254** are attached to the left and right panel structures **205***c*, **205***d* to provide the cabinet apparatus **100** with a clean aesthetic. As mentioned above, the second cover member **254** is shown flush with the outer wall of the right flange portion **253**. The cabinet apparatus **100** may be flush-mounted to the wall **500** using cleats, hanging hardware, screws, fasteners, mounting brackets, or the like.

[0059] FIG. **9** illustrates the cabinet apparatus **100** being recess-mounted to the wall **500**. When the cabinet apparatus **100** is recess-mounted to the wall **500**, a majority of the housing **200** is located within a hole formed into the wall **500** and a portion of the housing **200** and the door **300** are located outside of the hole in the wall **500**. In particular, the flange **260** abuts the outer surface of the wall **500** when the cabinet apparatus **100** is recess-mounted into the wall.

[0060] Referring to FIGS. **10** and **11** concurrently, when the cabinet apparatus **100** is recess-mounted to the wall **500**, the first and second cover members **244**, **254** are not attached to the housing **200**. This allows the flange **260** to abut against the wall to serve as a stopping point for the insertion of the housing **200** into the wall **500**. The flange **260** has a rear-most flange surface **270** that defines a wall contact plane B-B. The rear-most flange surface **270** may be formed by the distal end **266** of the abutment lip **263** of the left flange portion **243** (as well as by a distal end of an abutment lip of the second flange portion **253**). The rear-most flange surface **270** may also be formed by distal edges of the horizontal wall portion **228** of the top flange portion **221** and distal edges of the horizontal wall portion **235** of the bottom flange portion **209**.

[0061] The front access opening **299** of the storage space **202** lies in a first reference plane C-C. The first hinge component **280** protrudes from the front-outside corner **262** of the left flange portion **243** in a direction that is substantially parallel to the first reference plane C-C. The first reference plane C-C is also parallel to the wall contact plane B-B. Thus, the first hinge component **280** protrudes from the front-outside corner **262** of the left flange portion **243** in a direction that is substantially parallel to the wall contact plane B-B.

[0062] FIGS. **12** and **13** illustrate the same view as FIGS. **10** and **11**, except with the door **300** having been altered into the fully-open position. The hinge stop **335** of the door **300** and the stop lip **264** of the housing **200** are configured to interact with one another to prevent the door **300** from being rotated about the rotational axis A-A relative to the housing **200** beyond a predetermined angular position relative to the closed position, with the predetermined angular position being a fully open position of the door **300** relative to the housing **200**. The distal end **266** of the abutment lip **263** lies in a reference plane D-D which is substantially parallel to the reference plane C-C in which the front access opening **299** lies. The reference plane D-D may be the same plane as the wall contact plane B-B described previously. When the door **300** is in the fully-open position, no portion of the door **300** intersects the reference plane D-D. This ensures that no portion of the door **300** contacts, abuts, or scrapes the wall **500** when the door **300** is transitioned from the fully-closed position to the fully-open position. The door **300** may be unable to open beyond the full-open position due to the interaction between the hinge stop **335** and the stop lip **264**.

[0063] The predetermined angular position may be an angle $\Theta$**1** formed between the interior surface **302** of the door **300** and the front face **211** of the housing **200**, or may be the angle to which the door **300** is altered when moving from its fully-closed position to its fully-open position. The predetermined angular position may be greater than 90° from the closed position. The predetermined angular position may be at least 110° from the closed position.

[0064] While in the example shown the hinge stop **335** is located on the door **300** and abuts against the stop lip **264** on the housing **200** when the door **300** is in the fully-open position, alternative arrangements are possible. For example, in one alternative the hinge stop may be a feature located on the housing **200** which may abut or contact the door body **305** upon the door **300** being rotated into the fully-open position. In another alternative, the hinge stop **335** may remain on the door **300** but may contact part of the flange **260** directly rather than the stop lip **264** that protrudes from the flange **260**.

[0065] Referring to FIGS. **14** and **15**, a cabinet apparatus **600** is illustrated. The cabinet apparatus **600** includes a housing **601** and a door **602** that is pivotably coupled to the housing **601**. The features of the cabinet apparatus **100** described above are entirely applicable to the cabinet apparatus **600**. The only substantive difference, other than the size of the cabinet apparatus **600** relative to the cabinet apparatus **100**, is that the door **602** is round instead of rectangular. However, all of the other features including those regarding the hinge components, the flange and flange features, and the like are included in the cabinet apparatus **600**. The housing **601** may have an extension **604** along the left side where the hinge component is located to better enable the door **602** to fully close the access opening of the housing **601** when in the closed position. Otherwise, the features of the cabinet apparatus **600** are essentially the same as the cabinet apparatus **100** described above.

[0066] While the invention has been described with respect to specific examples including presently preferred modes of carrying out the invention, those skilled in the art will appreciate that there are numerous variations and permutations of the above-described systems and techniques. It is to be understood that other embodiments may be utilized and structural and functional modifications may be made without departing from the scope of the present invention. Thus, the spirit and scope of the invention should be construed broadly as set forth in the appended claims.

## Claims

**1**. A cabinet apparatus comprising: a housing defining a storage space having a front access opening, the housing comprising: a rear panel; a first panel structure including a first sidewall panel structure and a tubular flange extending lengthwise along the first sidewall panel structure, the tubular flange including a front wall, an outer wall, a rear wall, and an inner wall that collectively define an tubular cavity; and a first hinge component protruding from the tubular flange; and a door including a door body and a second hinge component on the door body, wherein the first and second hinge components are operably coupled to one another to so that the door is pivotably mounted to the housing to be rotatable about a rotational axis between: (1) a closed position in which the door body encloses the front access opening; and (2) an open position in which the storage space is accessible via the front access opening.

**2**. The cabinet apparatus of claim 1, wherein: the second hinge component includes a hinge stop; the front wall extends beyond the outer wall to form a stop lip that protrudes from an exterior surface of the outer wall; and the hinge stop and the stop lip are configured to interact with one another to prevent the door from being rotated about the rotational axis relative to the housing beyond a predetermined angular position relative to the closed position, the predetermined angular position being a fully open position of the door relative to the housing.

**3**. The cabinet apparatus of claim 1, wherein the first sidewall panel structure forms the inner wall of the tubular flange.

**4**. The cabinet apparatus of claim 1, wherein the first hinge component protrudes from a front-outside corner of the tubular flange.

**5**. The cabinet apparatus of claim 1, wherein the outer wall extends beyond the rear wall to form an abutment lip protruding from an exterior surface of the rear wall, the abutment lip abutting a wall surface when the cabinet apparatus is recess mounted into the wall.

**6**. The cabinet apparatus of claim 5, wherein when the door is in a fully-open position, no portion of the door intersects a second reference plane in which a distal edge surface of the abutment lip lies, the second reference plane being substantially parallel to a first reference plane in which the front access opening lies.

**7**. The cabinet apparatus of claim 6, wherein the abutment lip includes a distal end which abuts the wall surface when the cabinet apparatus is recess mounted in the wall, the distal end having an increased thickness relative to a remainder of the abutment lip.

**8**. The cabinet apparatus of claim 1, wherein the housing includes a magnetic element disposed on a front face of the housing, and wherein an interior surface of the door includes a pocket, the magnetic element disposed in the pocket when the door is in the closed position.

**9**. The cabinet apparatus of claim 1, further comprising: a top panel that forms an upper boundary of the storage space; and a hanging bracket extending upward from an exterior surface of the top panel.

**10**. A cabinet apparatus comprising: a housing defining a storage space having a front access opening, the housing comprising: a rear panel; a first panel structure including a first sidewall panel structure and a tubular flange extending lengthwise along the first sidewall panel structure, the tubular flange including a front wall, an outer wall, a rear wall, and an inner wall that collectively define a tubular cavity, the outer wall extending beyond the rear wall to form an abutment lip protruding from an exterior surface of the rear wall; and a first hinge component protruding from the tubular flange; and a door including a door body and a second hinge component on the door body, wherein the first and second hinge components are operably coupled to one another to so that the door is pivotably mounted to the housing.
**11**. The cabinet apparatus of claim 10, further comprising: a cover member detachably coupled to the first panel structure, the cover member abutting the abutment lip.
**12**. The cabinet apparatus of claim 11, wherein: the abutment lip includes a distal end having an increased thickness relative to a remainder of the abutment lip; an edge of the cover member abuts a distal end of the abutment lip; and a surface of the cover member is flush with an exterior surface of the outer wall of the tubular flange.
**13**. The cabinet apparatus of claim 10, wherein the first sidewall panel structure forms the inner wall of the tubular flange.
**14**. The cabinet apparatus of claim 10, further comprising: a top panel that forms an upper boundary of the storage space; and a hanging bracket extending upward from an exterior surface of the top panel.
**15**. A cabinet apparatus comprising: a housing defining a storage space having a front access opening, the housing comprising: a rear panel; a first panel structure including a first sidewall panel structure and a tubular flange extending lengthwise along the first sidewall panel structure; and a first hinge component protruding from the tubular flange; and a door including a door body and a second hinge component on the door body, wherein the first and second hinge components are operably coupled to one another to so that the door is pivotably mounted to the housing.
**16**. The cabinet apparatus of claim 15, wherein: the second hinge component includes a hinge stop; a front wall of the tubular flange extends beyond an outer wall of the tubular flange to form a stop lip that protrudes from an exterior surface of the outer wall; and the hinge stop and the stop lip are configured to interact with one another to prevent the door from being rotated about a rotational axis relative to the housing beyond a predetermined angular position.
**17**. The cabinet apparatus of claim 15, wherein the tubular flange includes a front wall, an outer wall, a rear wall, and an inner wall that collectively define a tubular cavity.
**18**. The cabinet apparatus of claim 17, wherein the first sidewall panel structure forms the inner wall of the tubular flange.
**19**. The cabinet apparatus of claim 15, wherein the housing includes a magnetic element disposed on a front face of the housing, and wherein an interior surface of the door includes a pocket, the magnetic element disposed in the pocket when the door is in a closed position.
**20**. The cabinet apparatus of claim 15, further comprising: a cover member detachably coupled to the first panel structure.